Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 29)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

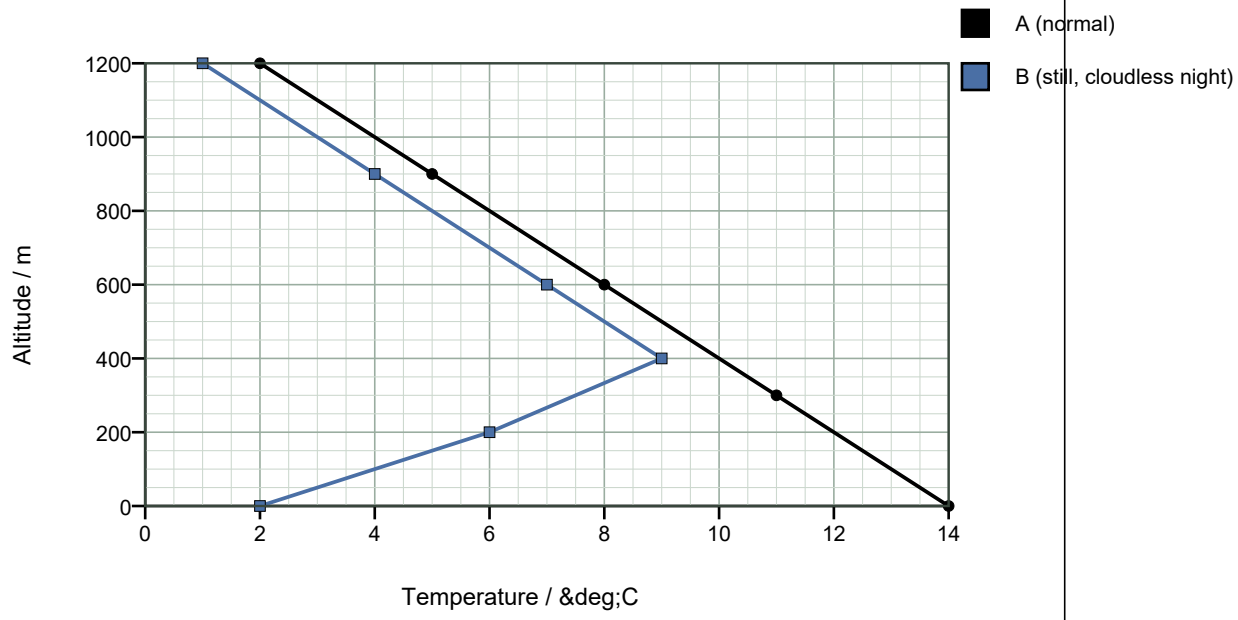
Question	Mark
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Answer all questions in the spaces provided.

0 1 Figure 1 shows how air temperature changes with altitude above a city in a valley on two different nights. Line A shows normal conditions. Line B shows the conditions on a still, cloudless night.

Figure 1



0 1 . 1 Name the atmospheric condition shown by line B in Figure 1.

[1 mark]

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0 1 . 2 Use Figure 1 to explain why pollutants released at ground level would disperse on night A but not on night B.

[4 marks]

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0 1 . 3 Explain why the conditions on night B (still, cloudless) allowed an inversion to form.

[3 marks]

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0 1 . 4 Give **two** types of pollution event that are made more severe by temperature inversions.

[2 marks]

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0 2 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 2 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 2 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

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0 2 . 3 Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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0 3 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 3 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 3 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

Answer

0 3 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

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0 3 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

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0 3 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

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0 4 Table 1 shows the mass of carbon in the main reservoirs of the carbon cycle.

Table 1

Reservoir	Mass of carbon / billion tonnes
Atmosphere (CO_2)	700
Living organisms	470
Dead organic matter (soil)	3 700
Surface ocean waters	500
Deep ocean (dissolved)	35 000
Fossil fuels	10 000
Sedimentary carbonate rocks	20 000 000

0 4 . 1 Express the mass of carbon in the atmosphere in standard form, in tonnes.

[1 mark]

Answer ☐ tonnes

0 4 . 2 Calculate the percentage increase in atmospheric carbon that would result from burning 10% of the fossil fuel reservoir, assuming all the carbon remained in the atmosphere.

[2 marks]

Answer ☐ %

0 4 . 3 Explain why the sedimentary carbonate rocks, although by far the largest reservoir, have little effect on atmospheric CO_2 concentrations.

[2 marks]

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0 4 . 4 Describe how carbon moves from the atmosphere into sedimentary carbonate rocks.

[3 marks]

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0 4 . 5 Explain why aerobic respiration returns carbon to the atmosphere as CO_2 but anaerobic respiration returns it mainly as methane.

[2 marks]

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0 5 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg^{-1}	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

0 5 . 1 Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

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0 5 . 2 Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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0 5 . 3 Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

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0 5 . 4 Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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0 6 A sand and gravel pit near a town is due to close. Three options for the site are proposed: a wetland nature reserve, housing, or return to farmland.

0 6 . 1 Explain why flooded sand and gravel pits are often developed as wetland nature reserves.

[2 marks]

0 6 . 2 Explain the conditions that would need to be met for the site to be used for housing or farmland.

[2 marks]

0 6 . 3 Explain why moving the wildlife from a site before mining begins is rarely completely successful.

[1 mark]

5

0 7 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 7 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 7 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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0 8 . 1 Which of the following statements about stratospheric ozone is correct? Tick one box.

[1 mark]

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☒

D Ozone concentrations are highest at ground level ☐

0 8 . 2 Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

[2 marks]

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0 8 . 3 Explain why HCFCs cause less ozone depletion than CFCs.

[2 marks]

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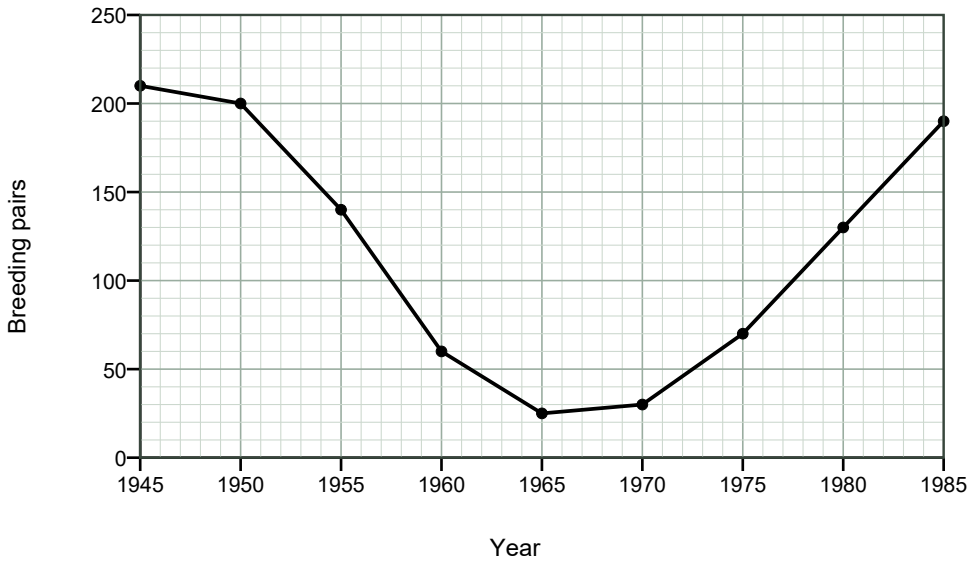
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0 9 Figure 1 shows the number of breeding pairs of sparrowhawks in a study area in the UK between 1945 and 1985, and the years in which DDT use began and was banned.

Figure 1



0 9 . 1 Describe the trend shown in Figure 1.

[2 marks]

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0 9 . 2 Calculate the percentage decrease in breeding pairs between 1950 and 1965. Show your working.

[2 marks]

Answer ____ %

0 9 . 3 DDT use in the UK began in the late 1940s and was restricted from the mid 1960s. Explain how the properties of DDT could account for the pattern in Figure 1.

[4 marks]

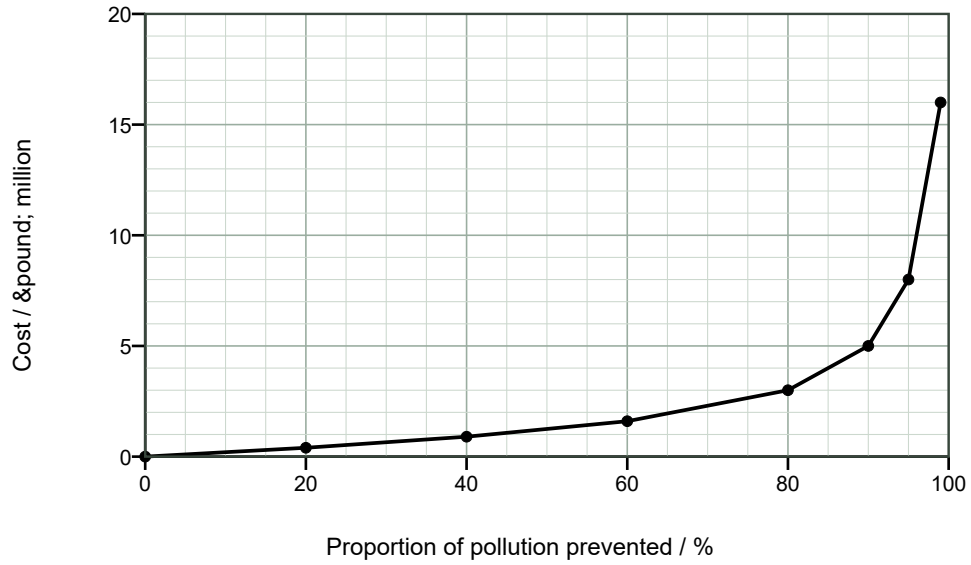
0 9 . 4 Suggest why the recovery after 1970 was slow.

[2 marks]

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1 0 Figure 1 shows how the cost of controlling emissions from a factory changes with the proportion of the pollutant that is prevented from being released.

Figure 1



1 0 . 1 Use Figure 1 to calculate the extra cost of increasing control from 90% to 99% of the pollutant.

[1 mark]

Answer £ million

1 0 . 2 Explain how the terms **ALARA** and **BATNEEC** are used when deciding how much pollution control is required.

[3 marks]

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1 0 . 3 Explain why the control methods that were originally adequate for a factory may no longer be sufficient.

[2 marks]

1 0 . 4 Evaluate the principles and general strategies that can be used to decide how pollution should be controlled.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

OR

1 1 . 2 To what extent can renewable energy resources replace fossil fuels in meeting the energy needs of a modern society?

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS